

MUNAZIL V

AI / Machine Learning Engineer

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SUMMARY

AI/ML engineer who has designed, built, and deployed multiple production multi-tenant SaaS platforms to live UAE clients within the first year of a full-stack career — owning backend, frontend, AI integration, and deployment end-to-end. Specialized in document intelligence and OCR (KYC, receipts, claims), LLM and multimodal AI pipelines, and AI-driven workflow automation. Comfortable across Python/FastAPI services, React frontends, Azure deployment, Docker, and CI/CD. Prior research experience at DRDO in multimodal retrieval.

EXPERIENCE

Machine Learning Developer

Aug 2025 – Present

Pure Magic IT Solutions — Dubai, UAE (Remote)

- Primary AI/ML engineer delivering production SaaS products end-to-end for real UAE clients — from architecture and backend to frontend, AI integration, Azure deployment, and CI/CD.
- Set up the team's GitHub organization with branch protection, push rules, and CI/CD pipelines across multiple projects; reviewed and merged pull requests and authored PRDs, deploy runbooks, and test-case documentation.
- Ran client demos and internal product training for sales teams; conducted security and QA audits (Playwright + Claude Code) across multiple production codebases, identifying and fixing vulnerabilities.

KEY PROJECTS

DocFoundry by Dexter AI — Multi-Tenant KYC & Document Verification Platform

- Built a multi-tenant identity-verification SaaS that ingests identity documents, runs **Azure Document Intelligence OCR** for field extraction, and verifies users via **InsightFace face recognition** — deployed to production for multiple client organizations.
- Engineered the full backend: JWT auth with refresh tokens, RBAC middleware (4 roles), document upload and storage, purchase-order and reporting modules, and 11+ REST endpoints; reached 85% test coverage with a pytest suite.
- Dockerized the full stack (FastAPI + React + Celery + Redis + PostgreSQL), deployed to Azure VM behind Nginx, and configured GitHub-to-server CI/CD for automated deployment; added a bulk-upload pipeline for high-volume processing.

Stack: Python, FastAPI, React, PostgreSQL, Celery, Redis, Azure Document Intelligence, InsightFace, Docker, Nginx, GitHub Actions CI/CD, JWT

Receipt OCR by Dexter AI — Receipt Digitization SaaS

- Built a multi-tenant receipt-digitization platform that extracts structured fields from receipt images and exports to Google Sheets, deployed to production with a mapped domain and used by a paying client.
- Solved multi-receipt detection in a single image through several iterations — from OpenCV contour and projection methods to a final **Azure prebuilt-layout** approach that returns receipt boundaries and coordinates for accurate content-aware auto-cropping.
- Designed an Azure OCR + OpenAI (GPT-4.1) extraction-and-mapping pipeline for field normalization; implemented JWT auth with OTP, “remember-me” background token refresh, real-time processing, and multi-worker Celery for concurrent jobs.

Stack: Python, FastAPI, React, PostgreSQL, Celery, Redis, Azure Document Intelligence, OpenAI GPT-4.1, Google Sheets API, Docker, Nginx, Azure VM

WorkTrack by Dexter AI — AI Event-Staffing Platform

- Designed and shipped a multi-tenant staffing platform that auto-assigns staff to events using an **OpenAI-powered recommendation engine** that ranks employees by skill and availability — replacing a manual spreadsheet workflow.
- Built the complete backend — auth, RBAC, staff/event/assignment modules, super-admin with multi-organization management and admin onboarding, and notifications — and drove the frontend integration to production.
- Deployed to Azure with GitHub-synced auto-deploy, daily database backups (7-day retention cron), firewall hardening, and tenant-isolation fixes; reduced a polling workflow latency from 10 minutes to 10 seconds.

Stack: Python, FastAPI, React, PostgreSQL, Alembic, OpenAI, Airtable, Docker, Azure VM, Nginx, GitHub Actions CI/CD

ClaimAssist — Insurance Claim Intake & AI Damage Assessment

- Built an automated claims pipeline that intakes claims via Email/Telegram/WhatsApp and live mobile-camera capture, then runs a multi-model AI assessment (**OpenAI GPT for images, Gemini 2.5 Pro for 360° video**) for damage severity, cost estimation, and fraud detection.
- Implemented OCR validation of number-plate and VIN against policy records, confidence-based routing to a human-review queue (low-confidence/severe claims), real-time Slack/email alerts, and a tamper-resistant audit log; auto-generated PDF claim summaries.

Stack: n8n, OpenAI, Google Gemini 2.5 Pro, Azure Blob Storage, PostgreSQL (Neon), OCR, APITemplate.io, Slack/Email integrations

AEO Audit — AI-Powered Website Audit Tool

- Built and deployed a full-stack web tool (frontend + backend, mapped to a live subdomain) that generates AI-driven AEO/website audit reports; integrated Calendly booking and onboarded real inbound leads.

Stack: Python, FastAPI, React, Claude API, Nginx, Azure VM

Additional AI & Automation Projects

- **Lead-Management CRM & Conversational Bots:** AI lead-qualification chatbots and CRM funnels (GoHighLevel + n8n) with Bayut/Property Finder integration; restaurant ordering bot with menu, price breakdown, table booking, and Google Calendar sync.
- **Content Intelligence Engine:** Reddit/X scraping workflow that surfaces trending topics and auto-generates hooks, captions, and hashtags for a queried subject.
- **Event Template Creator (Google Workspace Add-on):** Built a Google Calendar add-on in Apps Script with dynamic role/skill fields and event ranking, published live and submitted to Google for production verification.
- **Maritime CLIP Search (DRDO research):** CLIP-based semantic image-caption retrieval for maritime surveillance; improved query-image match accuracy by 15% via feature fusion and WordNet synonym expansion.
github.com/Munazil1/maritime-clip-search

RESEARCH EXPERIENCE

AI & ML Intern

Apr 2025 – Jul 2025

Centre for Air Borne Systems (CABS), DRDO — Bengaluru, India

- Built a semantic image-caption search system for maritime surveillance using OpenAI CLIP and PyTorch, designing embedding pipelines and re-ranking models for multimodal retrieval.
- Improved query-image match accuracy by 15% through feature-fusion techniques and WordNet synonym expansion; targeted offline-capable deployment for air-gapped environments.

EDUCATION

B.E. in Artificial Intelligence and Machine Learning

2021 – 2025

Visvesvaraya Technological University, SEACET, Bengaluru | CGPA: 7.6

Higher Secondary Education (Biology / Maths / Physics / Chemistry)

2019 – 2021

CHM Higher Secondary School, Elayavoor | 93.5%

Secondary School Leaving Certificate (SSLC)

2018 – 2019

CHM Higher Secondary School, Elayavoor | 100%

PUBLICATION & RECOGNITION

- **“Voice-Based Email for Visually Impaired People”** — presented at PRAVARTANA’24, S.E.A. College of Engineering in collaboration with IEEE Bangalore Section (Dec 2024).
- AI & ML Internship Certificate — DRDO Centre for Air Borne Systems (CABS), 2025.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, HTML/CSS

AI / ML: OpenAI (GPT-4o/4.1), Google Gemini 2.5 Pro, Claude API, LangChain, CLIP, PyTorch, InsightFace, Azure Document Intelligence, OCR, multimodal & RAG pipelines, prompt engineering, human-in-the-loop AI

Backend: FastAPI, REST API design, JWT/OTP auth, RBAC, Celery, Redis, SQLAlchemy, Alembic, PostgreSQL, multi-tenant architecture, WebSockets

Frontend: React, React Router, Lovable, responsive UI, dashboard development

Cloud & DevOps: Microsoft Azure (VM, Blob Storage, Form Recognizer), Docker, Docker Compose, Nginx, GitHub Actions CI/CD, Linux server administration, SSH deployment, database backups

Automation & Tools: n8n, GoHighLevel, Airtable, Google Apps Script, Google Sheets/Calendar APIs, Git/GitHub, Claude Code, Playwright

Spoken Languages: English (fluent), Malayalam (native), Tamil (fluent), Hindi (intermediate), Arabic (basic)